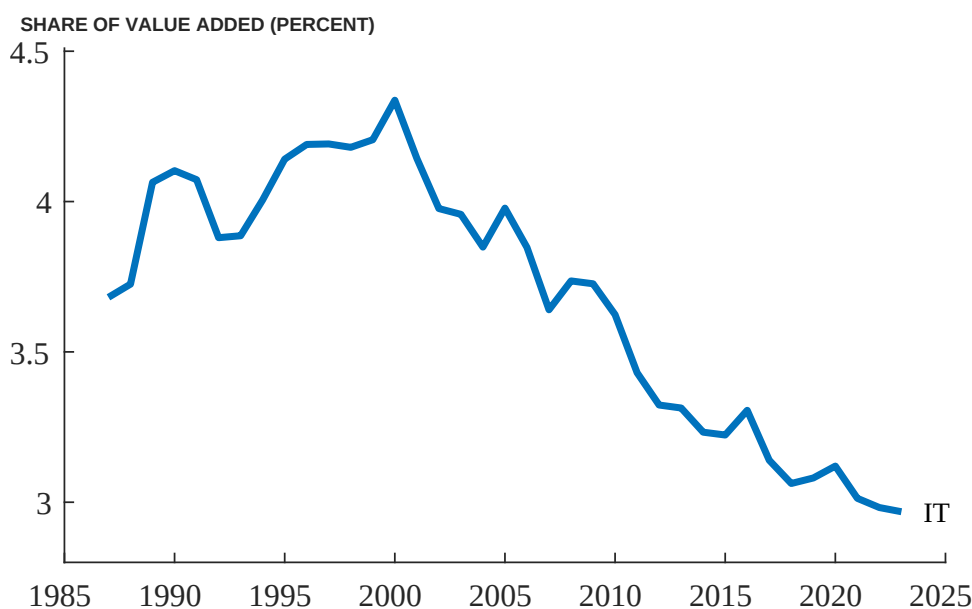


Better computers and weak links

Computerization, factor rewards, and output

Faster computers allow more calculations, searches, and records to be processed with the same physical equipment. It might seem natural to expect an increasingly productive technology to receive an increasing share of the economy's income. Yet the U.S. share of private-business value added paid to information technology (IT) fell after its peak around 2000. Exhibit 1 documents that pattern. Exhibit 2 shows another feature of the same broad technological era: productivity grew much faster in computer and electronic products than in several other industries. How can rapid productivity growth in computer production coexist with a falling income share for IT capital? How does progress in one activity affect total production and the payments for different inputs?

Exhibit 1. The income share paid to information technology



Historical evidence. U.S. private business: payments for IT capital services as a share of value added.

Exhibit 2. Uneven productivity growth across industries

U.S. industry	Average annual TFP growth, 1987–2017
Computers and electronic products	12.8%
Motor vehicles	1.7%
Retail trade	1.7%
Publishing (including software)	1.8%

Historical evidence. U.S. industry TFP growth, measured as average annual log changes over 1987–2017.

IT capital income comes from payments for capital services; it differs from purchases of new computers and from computer manufacturers' sales, which are recorded into aggregate investment. There is also a distinction between a computer's technical capability and the value of an additional unit of its services. Once routine information is abundant, producing still more of it may add little unless other activities expand as well. For example, a hospital can process records faster without providing more bedside assistance. A building project can generate more designs without completing more wiring.

Earlier waves of automation changed which activities machines could perform. Combine harvesters replaced workers threshing grain by hand; automated systems took over much elevator operation, typing, and travel booking; and industrial robots took over spot welding and spray painting in car plants. These changes brought more activities within the reach of improving machines. As machines improved, output rose reflecting the higher productivity of the activities they performed. Yet, some activities remained out of reach for machines, and the relative scarcity of those activities limited the overall gains from computerization.

The economic issue is *substitutability*. If additional computing services are a poor substitute for the remaining activities, progress concentrated in computing leaves those activities relatively scarce. Production can increase, but it may become increasingly constrained by what has not improved. The term *weak link* describes this: an activity matters because production still needs it, it is relatively weak in the chain of production (but that does not mean it is unproductive in an absolute sense).

Consider a capital–labour CES production function:

$$Y = z \left[\alpha (A_K K)^{(\sigma-1)/\sigma} + (1 - \alpha) (A_L L)^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad \sigma > 0, \sigma \neq 1. \quad (1)$$

Here K and L are physical capital and labour services; A_K and A_L are their respective productivity terms; z is common productivity; and α is a fixed technology weight. At $\sigma = 1$, use the Cobb–Douglas limit. An improvement in computing raises A_K while K , L , A_L , z , α , and the assignment of activities remain fixed in Questions 1–3.

Capital and labour are fully employed, and competitive real factor rewards equal their marginal products. Here p is the output price, w is the payment per unit of labour, and r is the rental payment per unit of capital.

With competitive production, the real payment per unit of each input equals its marginal product:

$$\frac{r}{p} = z^{(\sigma-1)/\sigma} \alpha A_K^{(\sigma-1)/\sigma} \left(\frac{Y}{K} \right)^{1/\sigma}, \quad \frac{w}{p} = z^{(\sigma-1)/\sigma} (1 - \alpha) A_L^{(\sigma-1)/\sigma} \left(\frac{Y}{L} \right)^{1/\sigma}.$$

The income shares and relative factor reward satisfy

$$\alpha_K = \frac{rK}{pY}, \quad \alpha_L = \frac{wL}{pY}, \quad \alpha_K + \alpha_L = 1, \quad \frac{\alpha_K}{\alpha_L} = \frac{\alpha}{1 - \alpha} \left(\frac{A_K K}{A_L L} \right)^{(\sigma-1)/\sigma}.$$

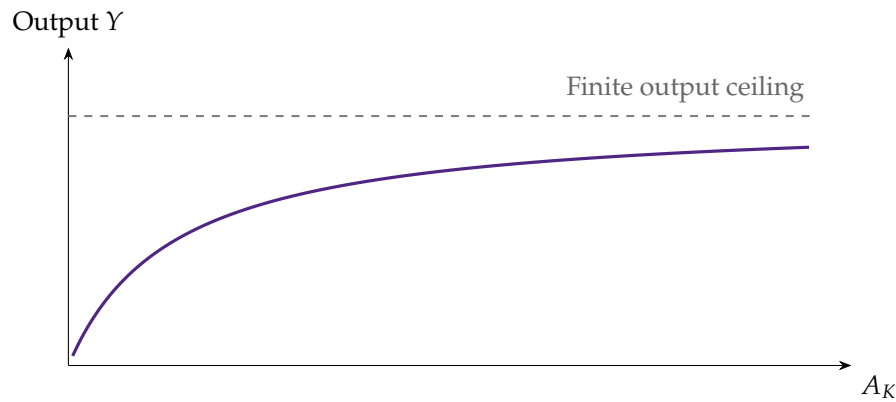
When only A_K changes with physical supplies and other technology fixed, it holds that:

$$\frac{\partial \log Y}{\partial \log A_K} = \alpha_K, \quad \frac{\partial \log(w/p)}{\partial \log A_K} = \frac{\alpha_K}{\sigma}, \quad \frac{\partial \log(r/p)}{\partial \log A_K} = 1 - \frac{\alpha_K}{\sigma}.$$

Exhibit 3. The weak-link limit

For $0 < \sigma < 1$ under the fixed-assignment experiment, the output ceiling and potential proportional gain from initial output Y_0 are

$$\bar{Y} \equiv \lim_{A_K \rightarrow \infty} Y = z(1 - \alpha)^{-\sigma/(1-\sigma)} A_L L, \quad \frac{\bar{Y}}{Y_0} = (1 - \alpha_{K,0})^{-\sigma/(1-\sigma)} = (\alpha_{L,0})^{-\sigma/(1-\sigma)}.$$



Question 1. What does the weak link limit?

(30 marks)

(a) A commentator claims that eventually a further improvement in computers has no value. Evaluate the claim using the CES model explaining what $\sigma < 1$ mean for substitution between capital and labour. Give a concrete production example and use it to explain the finite output limit in Exhibit 3. [20 marks]

(b) Using the second result in Exhibit 3, explain how the initial labour income share affects the potential gain from better computers. If two economies have the same initial output and σ but different initial labour shares, which has the larger gain and why? [10 marks]

Question 2. Can better machines receive a smaller income share?

(20 marks)

A manager claims that more productive computers must earn a larger share of income. Evaluate the claim using the CES model above. What happens to the two income shares and relative factor rewards? How does this depend on the elasticity of substitution? Explain why the result makes sense (you can give examples as part of the explanation).

Question 3. Do income shares reveal real rewards?

(25 marks)

An investor claims that a falling capital share proves that the real rental rate r/p falls. Under what conditions does r/p rise, fall, or remain unchanged as computer productivity increases? What happens to wages? Explain your answer.

Question 4. Does every computer technology face the same limit?

(15 marks)

A manager claims that Exhibit 3 misses the gains from computer technology. Evaluate the claim for faster performance of existing machine activities, a tool that helps workers at their existing activities, and machines taking over an activity previously performed by workers.

Question 5. What do the historical exhibits establish?

(10 marks)

A commentator claims that Exhibits 1 and 2 prove that capital and labour are poor substitutes and that Canada has little to gain from better computers. Evaluate the claim using the task assignment model and separate your analysis into what you think happens before and after the introduction of artificial intelligence technologies. You can use figures to illustrate your answer.

References

This case study is based on results and exhibits from the following paper. The questions and classroom illustration in Exhibit 3 were prepared for this course.

- Jones, Charles I., and Christopher Tonetti (2026). *Past Automation and Future A.I.: How Weak Links Tame the Growth Explosion*. March 29, version 0.4 (course copy).